

第八次习题课小练习

求解递推关系 $a_n = a_{n-1} + a_{n-2}$ ($for\ n > 1$),
其中 $a_0 = 1, a_1 = 2$.

解：构造生成函数 $G(x) = \sum_{i=0}^{\infty} a_i x^i$, 则有

$$(1 - x - x^2)G(x) = x + 1, G(x) = \frac{x+1}{1-x-x^2}$$

下面分解 $G(x)$: 设 $(x - a)(x - b) = x^2 + x - 1$,

$$\text{有 } ab = -1, a + b = -1 \Rightarrow a^2 + a - 1 = 0 \Rightarrow a = \frac{-1 \pm \sqrt{5}}{2},$$

$$\left(x - \frac{-1+\sqrt{5}}{2}\right)\left(x - \frac{-1-\sqrt{5}}{2}\right) = x^2 + x - 1.$$

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那么再设 $G(x) = \frac{c}{x - \frac{-1+\sqrt{5}}{2}} + \frac{d}{x - \frac{-1-\sqrt{5}}{2}}$, 对照原式 $G(x) = \frac{x+1}{1-x-x^2}$,

有 $c + d = -1, \frac{-1-\sqrt{5}}{2}c + \frac{-1+\sqrt{5}}{2}d = 1$,

解得 $c = \frac{-1-\frac{1}{\sqrt{5}}}{2}, d = \frac{-1+\frac{1}{\sqrt{5}}}{2}, G(x) = \frac{\frac{-1-\frac{1}{\sqrt{5}}}{2}}{x - \frac{-1+\sqrt{5}}{2}} + \frac{\frac{-1+\frac{1}{\sqrt{5}}}{2}}{x - \frac{-1-\sqrt{5}}{2}}$

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变换得到 $G(x) = \frac{\frac{\frac{3}{5}\sqrt{5}+1}{2}}{1-\frac{\sqrt{5}+1}{2}x} + \frac{\frac{-\frac{3}{5}\sqrt{5}+1}{2}}{1-\frac{-\sqrt{5}+1}{2}x}$, 代入生成函数公式可得

$$a_i = \frac{\frac{3}{5}\sqrt{5}+1}{2} \left(\frac{\sqrt{5}+1}{2} \right)^i + \frac{-\frac{3}{5}\sqrt{5}+1}{2} \left(\frac{-\sqrt{5}+1}{2} \right)^i$$